

Maxwell conjecture is false for four point charges

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Abstract

For distinct sites $a_i \in \mathbb{R}^3$ and nonzero charges q_i , let $V_{a,q}(x) = \sum_i q_i/|x - a_i|$, and let $M(n)$ be the supremum of the number of critical points over configurations of n charges whose critical set is finite. We prove that $M(4) \geq 11$. Four positive charges can have a finite critical set containing at least eleven points, all nondegenerate. This disproves Maxwell's conjectured bound of $M(4) \leq 9$. This is the smallest counterexample to Maxwell conjecture, since we proved $M(3) = 4$ in an earlier paper.

Our proof uses a bifurcation from a totally degenerate equilibrium of a trigonal pyramid: decreasing the apex charge produces eight nearby equilibria, and a global degree formula forces at least three more. The eight-point local bifurcation occurs when the apex height H above a unit-circumradius equilateral triangle satisfies $H > \sqrt{2}$.

1 Introduction

Given finitely many point charges in three-dimensional space, where can a test charge be in equilibrium, and how many such positions can occur? If V denotes the Coulomb potential, then the electric field is $-\nabla V$, so its equilibria are precisely the critical points of V . This elementary formulation leads to a subtle counting problem for a special class of harmonic functions with prescribed Coulomb singularities. Since V is nonconstant and harmonic away from the charges, the strong maximum principle rules out local maxima and minima there, but it gives no upper bound on the number of equilibria. When all charges have the same sign, Lemma 6.1 below, applied after an overall sign change if necessary, places every equilibrium in the relative interior of the convex hull of the sites; even so, the number and the local types of the equilibria can change under deformation.

We use the following notation throughout. Let $n \geq 1$, let $a = (a_1, \dots, a_n) \in (\mathbb{R}^3)^n$ have pairwise distinct entries, and let $q = (q_1, \dots, q_n) \in (\mathbb{R} \setminus \{0\})^n$. Put

$$\begin{aligned} \Omega_a &= \mathbb{R}^3 \setminus \{a_1, \dots, a_n\}, & V_{a,q} : \Omega_a &\longrightarrow \mathbb{R}, \\ V_{a,q}(x) &= \sum_{i=1}^n \frac{q_i}{|x - a_i|}, & \text{Crit}(V_{a,q}) &= \{x \in \Omega_a : \nabla V_{a,q}(x) = 0\}. \end{aligned}$$

We call (a, q) a *charge configuration* and the points of $\text{Crit}(V_{a,q})$ its *equilibria*. An equilibrium x is *nondegenerate* if $\det \text{Hess } V_{a,q}(x) \neq 0$ and *totally degenerate* if $\text{Hess } V_{a,q}(x) = 0$. A configuration is *Morse* if all of its equilibria are nondegenerate. Since some configurations may have nonisolated critical sets, we restrict the extremal count to configurations with finite critical set and define

$$M(n) = \sup \left\{ \# \text{Crit}(V_{a,q}) : \begin{array}{l} a \in (\mathbb{R}^3)^n, \ a_i \neq a_j \ (i \neq j), \\ q \in (\mathbb{R} \setminus \{0\})^n, \ \# \text{Crit}(V_{a,q}) < \infty \end{array} \right\},$$

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where critical points are counted as distinct real points, without multiplicity. This definition permits finite degenerate critical sets, whereas Maxwell's assertion concerns Morse configurations; the configuration in Theorem 1.1 is Morse.

In modern terminology, Maxwell's discussion is interpreted as asserting that a Morse configuration of n point charges has at most $(n-1)^2$ equilibria [8, Art. 113]; the gap in the original argument and the modern formulation are given by Gabrielov, Novikov, and Shapiro [4, Conjecture 1.3 and Appendix]. Morse and Cairns independently posed the problem [9, p. 293]. Gabrielov, Novikov, and Shapiro obtained the first bound depending only on n for the equilibria of a Morse configuration, using fewnomial methods. Zolotov later sharpened bounds for isolated equilibria via the Thom–Milnor theorem [11]. Edelsbrunner, Fillmore, and Oliveira obtained a further Bézout-type bound under a generic-position hypothesis and also constructed families with many equilibria [2].

The small cases illustrate the difficulty. For two equal positive charges, the midpoint is the unique equilibrium, so the proposed bound 1 is sharp. For three charges the proposed bound is 4, and three equal positive charges at the vertices of an equilateral triangle attain it [2]. Tsai proved the four-point upper bound for all triples of equal-magnitude charges [10]. A recent preprint reduces the best general upper bound on the number of nondegenerate equilibria of three positive charges from twelve to six [3, Theorem 1.1]. In a recent paper [6] we proved that $M(3) = 4$.

For four positive coplanar charges, Lee and Tsai constructed configurations with nine equilibria, attaining Maxwell's proposed value in that setting [7]. Another recent preprint constructs a Morse configuration of five charges with at least twenty-four equilibria, exceeding the proposed bound 16 [1]. The present result shows that failure already occurs for four charges, even when all charges are positive.

If $U \subset \mathbb{R}^3$ is open, $F : U \rightarrow \mathbb{R}^3$ is continuous, and $x \in U$ is an isolated zero of F , we write $\text{ind}_x(F)$ for its local Brouwer degree, also called its local Poincaré–Hopf index. Thus a nondegenerate critical point of V has $\text{ind}_x(\nabla V) = \text{sgn det Hess } V(x)$.

Theorem 1.1. *There are four affinely independent sites and four positive charges for which the critical set of the corresponding potential $V_{a,q}$ is finite, every critical point is nondegenerate, and*

$$\# \text{Crit}(V_{a,q}) \geq 11 > (4-1)^2 = 9.$$

Consequently Maxwell's proposed bound fails for $n = 4$, and $M(4) \geq 11$.

Our proof combines a local bifurcation with a global index constraint. Sections 2 and 3 construct an axial equilibrium with vanishing Hessian in a symmetric trigonal-pyramid family and compute its homogeneous cubic Taylor term exactly. Sections 4 and 5 analyze the resulting local model, show that lowering the apex charge creates at least eight nearby nondegenerate equilibria, and prove that the Brouwer degree on a fixed neighborhood is zero. Section 6 confines all equilibria, perturbs the charge vector arbitrarily slightly within the positive orthant to obtain a Morse configuration, and uses the global degree formula to force at least three additional equilibria.

2 The degenerate seed

We carry out the bifurcation analysis in the following C_{3v} -invariant two-parameter family, which we call the *trigonal pyramid*. For $H > 0$, fix

$$b_0 = (1, 0, 0), \quad b_1 = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right), \quad b_2 = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right), \quad b_3 = (0, 0, H), \quad (1)$$

assign charges $q_0 = q_1 = q_2 = 1$ and $q_3 = p > 0$, and denote the corresponding potential by

$$V_{H,p}(x) = \sum_{j=0}^2 \frac{1}{|x - b_j|} + \frac{p}{|x - b_3|}.$$

We write V when H and p are understood. The configuration is invariant under $C_{3v} \cong S_3$, the order-six group generated by rotation through $2\pi/3$ about the z -axis and reflection across the plane $y = 0$. Lemma 2.1 identifies a one-parameter curve in (H, p) -space on which an axial equilibrium is totally degenerate. The final generic perturbation of the charge vector need not preserve this symmetry.

Lemma 2.1 (Degenerate family). *Let $0 < t < 1/\sqrt{2}$ and put*

$$H = \frac{3t}{1 - 2t^2}, \quad p_0 = \frac{12t^3\sqrt{1+t^2}}{(1 - 2t^2)^2} > 0. \quad (2)$$

Then the point $x_0 = (0, 0, t)$ is a totally degenerate equilibrium of V_{H,p_0} ; that is,

$$\nabla V_{H,p_0}(x_0) = 0 \quad \text{and} \quad \text{Hess } V_{H,p_0}(x_0) = 0.$$

Proof. Put $d = 1 - 2t^2 \in (0, 1)$. Since $H/t = 3/d > 1$, we have $0 < t < H$, so x_0 lies in the domain of the potential: it differs from the apex because $t < H$, and from every base site because $t > 0$. Every point of the z -axis is fixed by C_{3v} , which acts on $\mathbb{R}^3$ as the direct sum of the trivial representation on $\mathbb{R}e_z$ and the standard two-dimensional irreducible real representation of S_3 on the horizontal plane. At such a point, equivariance gives $\nabla V = ce_z$ and $\text{Hess } V = \text{diag}(\lambda, \lambda, \mu)$, the latter by Schur's lemma. Since V is harmonic away from the sites, $2\lambda + \mu = \text{tr Hess } V = 0$. Consequently $c = 0$ and $\mu = 0$ are equivalent to $\nabla V(x_0) = 0$ and $\text{Hess } V(x_0) = 0$, respectively. Thus x_0 is a totally degenerate equilibrium if and only if

$$\partial_z V(x_0) = 0 \quad \text{and} \quad \partial_z^2 V(x_0) = 0.$$

Put $s = t - H$. At x_0 , the squared distance to each triangle charge is $1 + t^2$, and

$$\partial_z V(x_0) = -\frac{3t}{(1 + t^2)^{3/2}} - \frac{ps}{|s|^3}, \quad \partial_z^2 V(x_0) = \frac{3(2t^2 - 1)}{(1 + t^2)^{5/2}} + \frac{2p}{|s|^3}.$$

For the values in (2),

$$s = -\frac{2t(1 + t^2)}{d} < 0, \quad \frac{p_0}{|s|^2} = \frac{3t}{(1 + t^2)^{3/2}}, \quad \frac{2p_0}{|s|^3} = \frac{3d}{(1 + t^2)^{5/2}}.$$

Since $s < 0$ and $3(2t^2 - 1) = -3d$, these identities make both displayed derivatives vanish.

Conversely, suppose $t > 0$, $p > 0$, $s = t - H \neq 0$, and the two scalar equations above hold. The first forces $s < 0$, and the second then forces $t < 1/\sqrt{2}$. Eliminating p gives

$$s = \frac{2t(1 + t^2)}{2t^2 - 1},$$

after which $H = t - s$ and back-substitution for p give exactly (2). Thus these are the only solutions of the two scalar equations with $t > 0$ and $p > 0$. $\square$

Remark 2.2 (Axis formulation). There is a useful equivalent description of the degeneracy curve. For $0 < z < H$,

$$\partial_z V(0, 0, z) = \frac{p - g_H(z)}{(H - z)^2}, \quad g_H(z) := \frac{3z(H - z)^2}{(1 + z^2)^{3/2}},$$

and symmetry makes the other two components of the gradient vanish. Thus the axis equilibria in $(0, H)$ are exactly the solutions of $p = g_H(z)$. For $z \leq 0$, the base-triangle and apex contributions to $\partial_z V(0, 0, z)$ have positive sum, whereas for $z > H$ both contributions are negative. Hence every axis equilibrium lies in $(0, H)$.

For $0 < z < H$,

$$\frac{g'_H(z)}{g_H(z)} = \frac{H - 3z - 2Hz^2}{z(H - z)(1 + z^2)}.$$

Let $N_H(z) = H - 3z - 2Hz^2$ denote the numerator. Since

$$N'_H(z) = -3 - 4Hz < 0, \quad N_H(0) = H > 0, \quad N_H(H) = -2H(1 + H^2) < 0,$$

it has a unique root in $(0, H)$, namely

$$z_*(H) = \frac{-3 + \sqrt{9 + 8H^2}}{4H}.$$

It follows that g_H increases strictly from 0 to $g_H(z_*)$ and then decreases strictly to 0. Consequently there are two, one, or zero axis equilibria according as $p < g_H(z_*)$, $p = g_H(z_*)$, or $p > g_H(z_*)$.

At an axis equilibrium, differentiating the displayed formula for $\partial_z V$ gives

$$\partial_z^2 V(0, 0, z) = -\frac{g'_H(z)}{(H - z)^2}.$$

The C_{3v} argument in the proof of Lemma 2.1 therefore shows that, for each fixed $H > 0$, the only $p > 0$ for which an axial equilibrium is totally degenerate is $p = p_*(H) := g_H(z_*(H))$, and the corresponding equilibrium is $z = z_*(H)$. The identity $N_H(t) = 0$ is equivalent to

$$H = \frac{3t}{1 - 2t^2}, \quad g_H(t) = \frac{12t^3\sqrt{1 + t^2}}{(1 - 2t^2)^2},$$

so $t = z_*(H)$ and $p_0 = p_*(H)$ recover exactly the parameterization (2). In particular, for $H = 3$ one has $z_* = \frac{1}{2}$ and $p_*(3) = g_3(\frac{1}{2}) = \frac{75/8}{5\sqrt{5/8}} = 3\sqrt{5}$.

At $t = \sqrt{2}/4$, formula (2) gives $H = \sqrt{2}$ and $p_0 = 1$. The three base edges and the three lateral edges all have length $\sqrt{3}$, so this is a regular tetrahedron with four equal charges. After translation to its centroid and uniform rescaling, it agrees with the normalization used in Example 6.4; its centroid is $x_0 = (0, 0, \sqrt{2}/4)$. The tetrahedral symmetry also gives a short independent explanation for the vanishing Hessian: its linear action about x_0 is irreducible on $\mathbb{R}^3$, so the invariant symmetric form $\text{Hess } V_{\sqrt{2}, 1}(x_0)$ is scalar, while harmonicity makes its trace zero.

3 The three-jet

Let $Y = (u, v, \zeta) = (x, y, z - t)$, so the symmetry axis is the ζ -axis and $v = 0$ is a mirror plane, and write $w = u + iv$ and $\omega = e^{2\pi i/3}$. The generators of C_{3v} act by $w \mapsto \omega w$ and $w \mapsto \bar{w}$, while fixing ζ . Rotation invariance in total degree three leaves ζ^3 , $\zeta w \bar{w}$, w^3 , and $\bar{w}^3$; reflection invariance therefore shows that every invariant real homogeneous cubic is a linear combination of

$$\zeta^3, \quad \zeta(u^2 + v^2), \quad \text{Re}(w^3) = u^3 - 3uv^2.$$

Since x_0 is fixed by C_{3v} and V_{H, p_0} is real analytic and harmonic near x_0 , each homogeneous Taylor term at x_0 is invariant and harmonic. Moreover,

$$\Delta(\zeta^3) = 6\zeta, \quad \Delta(\zeta(u^2 + v^2)) = 4\zeta, \quad \Delta \text{Re}(w^3) = 0.$$

Thus the invariant harmonic cubics form the two-dimensional space spanned by

$$\zeta(2\zeta^2 - 3u^2 - 3v^2), \quad u^3 - 3uv^2.$$

Because Lemma 2.1 gives $\nabla V_{H,p_0}(x_0) = 0$ and $\text{Hess } V_{H,p_0}(x_0) = 0$, there is a unique homogeneous cubic h such that

$$V_{H,p_0}(x_0 + Y) = V_{H,p_0}(x_0) + h(Y) + O(|Y|^4).$$

Thus the three-jet of V_{H,p_0} at x_0 is $V_{H,p_0}(x_0) + h$, and h is its cubic term.

Lemma 3.1 (Exact cubic term). *On the family of Lemma 2.1, the cubic Taylor term h is*

$$h(u, v, \zeta) = A \zeta(2\zeta^2 - 3u^2 - 3v^2) + B(u^3 - 3uv^2), \quad (3)$$

with

$$A = \frac{3(1 + 2t^2)}{8t(1 + t^2)^{7/2}}, \quad B = \frac{15}{8(1 + t^2)^{7/2}}. \quad (4)$$

Consequently

$$\frac{B}{A} = \frac{5t}{1 + 2t^2}, \quad \frac{B^2}{A^2} = \frac{25t^2}{(1 + 2t^2)^2} = \frac{25H^2}{8H^2 + 9}, \quad (5)$$

and therefore

$$\boxed{B^2 > 2A^2 \iff t^2 > \frac{1}{8} \iff H^2 > 2} \quad (6)$$

within the standing range $0 < t^2 < 1/2$.

Proof. Put $\varrho = \sqrt{1 + t^2}$ and $s = t - H < 0$. By the preceding classification, it suffices to determine A and B . Since $h(u, 0, 0) = Bu^3$, the number B is the coefficient of u^3 in the Taylor expansion of $u \mapsto V(x_0 + (u, 0, 0)) = V(u, 0, t)$ at $u = 0$. The apex contribution to $V(u, 0, t)$ is even in u and hence has no u^3 term. For a base charge at $(\cos \theta_k, \sin \theta_k, 0)$, put $c_k = \cos \theta_k$. Its distance from $(u, 0, t)$ has square $\varrho^2 - 2c_k u + u^2$, and

$$(\varrho^2 - 2c_k u + u^2)^{-1/2} = \frac{1}{\varrho} + \frac{c_k}{\varrho^3} u + \frac{3c_k^2 - \varrho^2}{2\varrho^5} u^2 + \frac{5c_k^3 - 3c_k \varrho^2}{2\varrho^7} u^3 + O(u^4).$$

Summing over $\theta_k = 0, \pm 2\pi/3$ and using $\sum_k c_k = 0$ and $\sum_k c_k^3 = 3/4$ gives

$$B = \frac{15}{8\varrho^7}.$$

Along the symmetry axis, the coefficient of ζ^3 is both $\partial_z^3 V(x_0)/6$ and $2A$; hence $A = \partial_z^3 V(x_0)/12$. Direct differentiation shows that the three base charges contribute $9t(3 - 2t^2)/(1 + t^2)^{7/2}$ to $\partial_z^3 V(x_0)$, while the apex charge contributes $-6p_0 \text{sgn}(s)/|s|^4$. Since $\text{sgn}(s)/|s| = 1/s$, the latter equals

$$-6 \frac{p_0}{|s|^3} \frac{1}{s}.$$

The degeneracy relations in Lemma 2.1 give

$$\frac{p_0}{|s|^3} = -\frac{3(2t^2 - 1)}{2(1 + t^2)^{5/2}}, \quad \frac{1}{s} = \frac{2t^2 - 1}{2t(1 + t^2)}.$$

Consequently the apex contribution is $9(2t^2 - 1)^2/(2t(1 + t^2)^{7/2})$, and

$$\partial_z^3 V(x_0) = \frac{9}{2t(1 + t^2)^{7/2}} [2t^2(3 - 2t^2) + (2t^2 - 1)^2] = \frac{9(2t^2 + 1)}{2t(1 + t^2)^{7/2}},$$

which proves (4). Dividing the two expressions in (4) gives B/A ; squaring and then substituting $H = 3t/(1 - 2t^2)$ gives the remaining identities in (5). Finally,

$$\frac{B^2}{A^2} - 2 = \frac{(8t^2 - 1)(2 - t^2)}{(1 + 2t^2)^2}, \quad \frac{25H^2}{8H^2 + 9} - 2 = \frac{9(H^2 - 2)}{8H^2 + 9}.$$

Because $0 < t^2 < 1/2$ and $H > 0$, these factorizations prove (6). $\square$

The distinguished member. For the subsequent construction, take $t = \frac{1}{2}$, which satisfies the strict inequality in (6). Then

$$H = 3, \quad p_0 = 3\sqrt{5}, \quad x_0 = (0, 0, \frac{1}{2}), \quad A = \frac{144\sqrt{5}}{625}, \quad B = \frac{48\sqrt{5}}{125}, \quad (7)$$

so $B/A = 5/3$ and

$$\kappa := 1 - \frac{2A^2}{B^2} = 1 - \frac{18}{25} = \frac{7}{25} > 0. \quad (8)$$

All scalar values and coordinates in (7) lie in $\mathbb{Q}(\sqrt{5})$, and the identities are exact.

4 The local model

Lemma 4.1 (Smoothness). *Suppose that $A, B \in \mathbb{R}$ and $(A, B) \neq (0, 0)$. The projective cubic $C_h = \{h = 0\} \subset \mathbb{P}^2(\mathbb{C})$ is singular if and only if $A = 0$, $B = 0$, or $B^2 = 2A^2$. In particular it is smooth for the member (7).*

Proof. Use homogeneous coordinates $[u : v : \zeta]$ on $\mathbb{P}^2(\mathbb{C})$. Direct differentiation gives

$$\nabla h = (-6A\zeta u + 3B(u^2 - v^2), -6v(A\zeta + Bu), 3A(2\zeta^2 - u^2 - v^2)).$$

Euler's identity

$$3h = uh_u + vh_v + \zeta h_\zeta$$

shows that every nonzero common zero of the three partial derivatives lies on C_h . Thus such common zeros are precisely the singular points of C_h .

First suppose $AB \neq 0$. If $v = 0$, then $h_u = 0$ gives either $u = 0$ or $Bu = 2A\zeta$, while $h_\zeta = 0$ gives $u^2 = 2\zeta^2$. The first alternative would give $u = \zeta = 0$, which is not a projective point. In the second alternative $u \neq 0$, since $u = 0$ would again force $\zeta = 0$ and hence the same contradiction; squaring $Bu = 2A\zeta$, substituting $u^2 = 2\zeta^2$, and cancelling u^2 therefore gives $B^2 = 2A^2$.

If $v \neq 0$, then $h_v = 0$ gives $A\zeta = -Bu$. Substitution into $h_u = 0$ gives $v^2 = 3u^2$; in particular $u \neq 0$. The equation $h_\zeta = 0$ then gives

$$\frac{2B^2u^2}{A^2} = u^2 + v^2 = 4u^2,$$

and hence again $B^2 = 2A^2$. Conversely, if $AB \neq 0$ and $B^2 = 2A^2$, then $[2A/B : 0 : 1]$ is a common zero of the three partial derivatives.

It remains to consider $AB = 0$. If $A = 0$ and $B \neq 0$, then $h = Bu(u^2 - 3v^2)$ is the union of three distinct lines through $[0 : 0 : 1]$ and is singular there. If $B = 0$ and $A \neq 0$, then $h = A\zeta(2\zeta^2 - 3u^2 - 3v^2)$ is the union of a line and a conic; they meet at $[1 : \pm i : 0]$, which are singular points of the union. This proves the classification. For (7), A and B are nonzero and $B^2/A^2 = 25/9 \neq 2$, so C_h is smooth. $\square$

Lemma 4.2 (Local model). *Assume $A, B \in \mathbb{R} \setminus \{0\}$, $\kappa = 1 - 2A^2/B^2 > 0$, and let $\gamma \in \mathbb{R}$ satisfy $\gamma/A < 0$. Put $e_\zeta = (0, 0, 1)$ and*

$$F_\gamma(Y) := \nabla h(Y) + \gamma e_\zeta, \quad Y = (u, v, \zeta). \quad (9)$$

Then $F_\gamma(Y) = 0$ has exactly eight solutions in $\mathbb{C}^3$. All eight are real and simple; four have Poincaré–Hopf index +1 and four have index -1. Explicitly, they are

$$2 \text{ axis points } (0, 0, \pm\zeta_1), \quad \zeta_1^2 = -\frac{\gamma}{6A}; \quad (10)$$

$$2 \text{ orbits of size 3 through } \left(\frac{2A\zeta}{B}, 0, \zeta\right), \quad \zeta \in \{\zeta_2, -\zeta_2\}, \quad \zeta_2 = \sqrt{-\frac{\gamma}{6A\kappa}}. \quad (11)$$

Proof. Since $\kappa > 0$, we have $B^2 > 2A^2$, so Lemma 4.1 shows that C_h is smooth. Homogenize with a new coordinate τ , and let $Z \subset \mathbb{P}^3(\mathbb{C})$ be the projective scheme cut out by the three quadrics

$$h_u = 0, \quad h_v = 0, \quad h_\zeta + \gamma\tau^2 = 0.$$

The intersection $Z \cap \{\tau = 0\}$ is empty: a point in this intersection would give a nonzero common zero of ∇h , and Euler's identity would make it a singular point of C_h . If Z had a positive-dimensional irreducible component, the projective dimension theorem would force that component to meet the hyperplane $\{\tau = 0\}$, a contradiction. Hence Z is a zero-dimensional complete intersection: the three quadrics meet properly in $\mathbb{P}^3$. Bézout's theorem therefore gives

$$\text{length}(Z) = 2 \cdot 2 \cdot 2 = 8.$$

Since Z has no point at infinity, the affine solution scheme has length 8; equivalently, the system has eight affine solutions counted with multiplicity.

If $v = 0$, the first equation reads $3u(Bu - 2A\zeta) = 0$. For $u = 0$ the third gives $6A\zeta^2 = -\gamma$, i.e. (10), real because $\gamma/A < 0$. For $u = 2A\zeta/B$ the third gives $6A\zeta^2 - 3A \cdot 4A^2\zeta^2/B^2 = -\gamma$, i.e. $6A\kappa\zeta^2 = -\gamma$, which is (11); it is real because $\kappa > 0$ and $\gamma/A < 0$.

If $v \neq 0$, the second equation gives $A\zeta = -Bu$, the first then gives $v^2 = 3u^2$, and the third gives $u^2 \cdot 6(B^2 - 2A^2)/A = -\gamma$, which has two real roots $\pm u_0$ since $B^2 - 2A^2 > 0$ and $\gamma/A < 0$; with $v = \pm\sqrt{3}u$ this is 4 points. These 4 points are exactly the nontrivial rotations of the two representatives in (11). Indeed, if $a = 2A\zeta/B$, the rotations of $(a, 0, \zeta)$ are $(-a/2, \pm\sqrt{3}a/2, \zeta)$; here $-a/2 = -A\zeta/B$, so they satisfy $A\zeta = -Bu$ and $v^2 = 3u^2$. Thus the eight solutions are $2 + 3 + 3$, all real and distinct.

Their local intersection multiplicities are positive and sum to 8, so each is 1. By the Jacobian criterion for this zero-dimensional complete intersection, this is equivalent to the nonsingularity of $DF_\gamma(Y) = \text{Hess } h(Y)$ at every solution. Hence all eight solutions are simple.

It remains to count their indices. By smoothness of C_h and Euler's identity, ∇h is nonzero on S^2 . The map

$$f : S^2 \longrightarrow S^2, \quad f(Y) = \frac{\nabla h(Y)}{|\nabla h(Y)|},$$

is even because ∇h is homogeneous of degree two. Hence $f = \bar{f} \circ \pi$, where $\pi : S^2 \rightarrow \mathbb{RP}^2$ is the quotient map. Since $H_2(\mathbb{RP}^2; \mathbb{Z}) = 0$, the induced map f_* on $H_2(-; \mathbb{Z})$ is zero, and therefore $\deg f = 0$.

Now put $m = \min_{|Y|=1} |\nabla h(Y)| > 0$. Choose R large enough that all eight zeros lie in B_R and $mR^2 > |\gamma|$. If $|Y| = R$ and $0 \leq \lambda \leq 1$, then homogeneity gives $|\nabla h(Y)| \geq mR^2$, and therefore

$$|\nabla h(Y) + \lambda\gamma e_\zeta| \geq mR^2 - |\gamma| > 0.$$

Thus F_γ is homotopic to ∇h through maps that do not vanish on ∂B_R . Homotopy invariance and additivity of Brouwer degree give

$$\sum_{F_\gamma(Y)=0} \text{ind}_Y(F_\gamma) = \deg(F_\gamma, B_R, 0) = \deg f = 0.$$

Every simple zero has index ± 1 ; because there are eight zeros and their indices sum to zero, four have each sign. $\square$

Remark 4.3 (The threshold). For the degenerate trigonal-pyramid family of Lemma 2.1, the strict inequality in Lemma 4.2 is essential. At the regular-tetrahedron boundary $\kappa = 0$, one has $\gamma \neq 0$, and the two nonaxial branches reduce to the inconsistent equation $0 = -\gamma$; only the two axial affine solutions remain. If $\kappa < 0$, then the nonaxial branch with $v = 0$ would require

$$\zeta^2 = -\frac{\gamma}{6A\kappa} < 0,$$

while the branch with $v \neq 0$ would require

$$u^2 = -\frac{\gamma A}{6(B^2 - 2A^2)} < 0.$$

Thus only the two axial solutions are real when $\kappa \leq 0$. Under the standing sign condition $\gamma/A < 0$, equation (6) shows that this family has eight real solutions precisely when $H^2 > 2$.

5 Persistence

Lemma 5.1 (Local index is zero). *Let x_0 be a totally degenerate equilibrium of a point-charge potential V , let $h \neq 0$ be its homogeneous cubic Taylor term, and suppose that the projective cubic curve $C_h = \{h = 0\} \subset \mathbb{P}^2(\mathbb{C})$ is smooth. Then there is $r_0 > 0$ such that x_0 is the only critical point of V in $\overline{B}(x_0, r_0)$ and $\text{ind}_{x_0}(\nabla V) = 0$.*

Proof. If $y \neq 0$ and $\nabla h(y) = 0$, Euler's identity gives $h(y) = 0$, so $[y]$ would be a singular point of C_h . Hence ∇h has no nonzero real zero, and

$$m := \min_{|y|=1} |\nabla h(y)| > 0.$$

Analyticity of V near x_0 and total degeneracy give constants $r_1 > 0$ and $C > 0$ with

$$\nabla V(x_0 + y) = \nabla h(y) + \mathcal{R}(y), \quad |\mathcal{R}(y)| \leq C|y|^3 \quad (|y| \leq r_1).$$

Put $d_0 = \min_i |x_0 - a_i| > 0$ and choose $r_0 > 0$ so small that

$$r_0 \leq r_1, \quad r_0 < \frac{d_0}{2}, \quad \text{and} \quad Cr_0 < \frac{m}{2}.$$

Since ∇h is homogeneous of degree two, for $0 < |y| \leq r_0$ one has

$$|\nabla V(x_0 + y)| \geq |y|^2(m - C|y|) \geq \frac{m}{2}|y|^2 > 0.$$

Thus x_0 is the only critical point in the closed ball.

For $0 \leq \lambda \leq 1$, set $\Psi_\lambda(y) = \nabla h(y) + \lambda \mathcal{R}(y)$. The same estimate shows that Ψ_λ is nonzero when $|y| = r_0$. After identifying $\partial B(0, r_0)$ with S^2 by $y = r_0 \omega$, homotopy invariance and the degree-two homogeneity of ∇h give

$$\text{ind}_{x_0}(\nabla V) = \deg\left(\frac{\nabla h}{|\nabla h|} : S^2 \longrightarrow S^2\right).$$

The normalized map $f = \nabla h/|\nabla h|$ is even because ∇h has degree two. Thus $f = \bar{f} \circ \pi$, where $\pi : S^2 \rightarrow \mathbb{RP}^2$ is the quotient map. Since $H_2(\mathbb{RP}^2; \mathbb{Z}) = 0$, one has $f_* = \bar{f}_* \circ \pi_* = 0$ on second homology, and hence $\deg f = 0$. $\square$

Lemma 5.2 (Persistence). *Fix $t > 0$ with $\frac{1}{8} < t^2 < \frac{1}{2}$, and let H, p_0 be given by (2). Put $x_0 = (0, 0, t)$, $s = t - H < 0$, and $V_* = V_{H, p_0}$. By Lemma 3.1, the coefficients A and B of the cubic Taylor term of V_* at x_0 are nonzero and satisfy $B^2 \neq 2A^2$, so Lemma 4.1 makes C_h smooth and Lemma 5.1 applies to V_* ; fix a corresponding radius r_0 .*

For $\varepsilon > 0$, let $V_\varepsilon = V_{H, p_0 - \varepsilon}$. There are $0 < \varepsilon_0 < p_0$ and $R > 0$ such that, for every $0 < \varepsilon < \varepsilon_0$, the potential V_ε has exactly eight critical points in $\overline{B}(x_0, R\sqrt{\varepsilon})$, and all eight are nondegenerate. Moreover, $R\sqrt{\varepsilon} < r_0$, the field ∇V_ε is nonzero on $\partial B(x_0, r_0)$, and

$$\deg(\nabla V_\varepsilon, B(x_0, r_0), 0) = 0. \tag{12}$$

Proof. Let $\phi(x) = |x - b_3|^{-1}$, so $V_\varepsilon = V_* - \varepsilon\phi$. Since $x_0 - b_3 = se_\zeta$ with $s < 0$,

$$\nabla\phi(x_0) = \frac{1}{s^2}e_\zeta, \quad \text{Hess}\phi(x_0) = \frac{1}{|s|^3} \text{diag}(-1, -1, 2).$$

The gradient and Hessian of V_* vanish at x_0 , and hence, exactly,

$$\nabla V_\varepsilon(x_0) = \gamma_\varepsilon e_\zeta, \quad \gamma_\varepsilon = -\frac{\varepsilon}{s^2}, \quad \text{Hess} V_\varepsilon(x_0) = \frac{\varepsilon}{|s|^3} \text{diag}(1, 1, -2).$$

Since $t > 0$, equation (4) gives $A > 0$, so $\gamma_\varepsilon/A < 0$.

Let h be the homogeneous cubic Taylor term of V_* at x_0 , and put $\delta = \sqrt{\varepsilon}$. For $\delta \neq 0$, define the rescaled field

$$G(Y, \delta) := \delta^{-2} \nabla V_{H, p_0 - \delta^2}(x_0 + \delta Y). \quad (13)$$

Because $V_{H, p_0 - \delta^2} = V_* - \delta^2\phi$, Taylor expansion gives

$$\delta^{-2} \nabla V_*(x_0 + \delta Y) = \nabla h(Y) + \delta S_0(Y, \delta),$$

and

$$-\nabla\phi(x_0 + \delta Y) = \hat{\gamma}e_\zeta + \delta S_1(Y, \delta), \quad \hat{\gamma} = -\frac{1}{s^2}.$$

Here $\eta \mapsto \nabla V_*(x_0 + \eta) - \nabla h(\eta)$ is real analytic near $\eta = 0$ and vanishes to order three, so $(Y, \delta) \mapsto \delta^{-3}(\nabla V_*(x_0 + \delta Y) - \delta^2 \nabla h(Y))$, which is real analytic for $\delta \neq 0$, extends real analytically across $\delta = 0$; the same division argument applies to $\nabla\phi(x_0 + \delta Y) - \nabla\phi(x_0)$, which vanishes to order one in δ . Thus S_0 and S_1 are real analytic wherever $x_0 + \delta Y$ avoids the sites, and (13) extends real analytically to $\delta = 0$ by setting

$$G(Y, \delta) = \nabla h(Y) + \hat{\gamma}e_\zeta + \delta S(Y, \delta), \quad S = S_0 + S_1. \quad (14)$$

By (6) we have $\kappa > 0$, and $\hat{\gamma}/A < 0$ because $\hat{\gamma} < 0 < A$. Lemma 4.2 therefore gives exactly eight real simple zeros $Y_1, \dots, Y_8$ of $G(\cdot, 0)$. Choose $R > 0$ and pairwise disjoint open neighborhoods U_j of the Y_j whose closures lie in $\{|Y| < R\}$. Let $d_0 = \min_i |x_0 - b_i| > 0$ and choose $\delta_0 > 0$ with $R\delta_0 < d_0/2$. Then S is analytic on an open neighborhood of the compact cylinder $\{|Y| \leq R\} \times [-\delta_0, \delta_0]$.

After shrinking the U_j if necessary, the implicit function theorem gives a unique real-analytic zero in each U_j . Taking the minimum of the eight parameter radii, there is $\delta_1 > 0$ such that these zeros define branches $Y_j(\delta) \in U_j$ for every $|\delta| < \delta_1$. Shrinking δ_1 if necessary, continuity also ensures that $D_Y G(Y_j(\delta), \delta)$ is invertible for every j and $|\delta| < \delta_1$. Let

$$E = \{|Y| \leq R\} \setminus \bigcup_{j=1}^8 U_j.$$

The compact set E contains no zero of $G(\cdot, 0)$, so $\min_E |G(Y, 0)| > 0$. Equation (14) gives uniform convergence $G(\cdot, \delta) \rightarrow G(\cdot, 0)$ on E . After shrinking the parameter interval, $G(\cdot, \delta)$ therefore has no zero on E . Hence the eight branches $Y_j(\delta)$ are exactly the zeros of $G(\cdot, \delta)$ in $\{|Y| \leq R\}$.

For $\delta > 0$,

$$D_Y G(Y, \delta) = \delta^{-1} \text{Hess}_x V_{H, p_0 - \delta^2}(x_0 + \delta Y).$$

Thus simplicity of the continued zeros implies that the corresponding critical points $x_j(\varepsilon) = x_0 + \sqrt{\varepsilon} Y_j(\sqrt{\varepsilon})$ of V_ε are nondegenerate. The scaling is a bijection between zeros of $G(\cdot, \sqrt{\varepsilon})$ in $\{|Y| \leq R\}$ and critical points of V_ε in $\overline{B}(x_0, R\sqrt{\varepsilon})$, proving the exact count.

Finally, set $\Sigma = \partial B(x_0, r_0)$. Since the seed field is nonzero on Σ ,

$$\mu := \min_{\Sigma} |\nabla V_*| > 0, \quad M_0 := \max_{\Sigma} |\nabla\phi| < \infty.$$

For $0 \leq \lambda \leq 1$,

$$|\nabla(V_* - \lambda \varepsilon \phi)(x)| \geq \mu - \varepsilon M_0 \quad (x \in \Sigma).$$

Thus, if $\varepsilon M_0 < \mu$, the straight charge homotopy is nonzero on Σ . Lemma 5.1 gives

$$\deg(\nabla V_*, B(x_0, r_0), 0) = \text{ind}_{x_0}(\nabla V_*) = 0,$$

so homotopy invariance gives (12). Shrinking ε_0 once more ensures $\varepsilon_0 < p_0$, $R\sqrt{\varepsilon_0} < r_0$, and all the preceding smallness conditions. $\square$

6 Global degree and the final perturbation

We first record two global statements: a confinement lemma for charges of one sign, and the global index identity, including the signs needed below.

Lemma 6.1 (Confinement). *If all $q_i > 0$, then every equilibrium lies in $\text{relint conv}\{a_i\}$. Moreover, fix a site a_i , let $\delta > 0$, and let numbers L_{ik} satisfy $\delta < L_{ik} \leq |a_i - a_k|$ for every $k \neq i$. If*

$$\frac{q_i}{\delta^2} > \sum_{k \neq i} \frac{q_k}{(L_{ik} - \delta)^2}, \quad (15)$$

then $\nabla V(x) \neq 0$ whenever $0 < |x - a_i| \leq \delta$.

Proof. The equilibrium equation reads

$$\sum_i w_i(a_i - x) = 0, \quad w_i = \frac{q_i}{|x - a_i|^3} > 0.$$

After division by $\sum_i w_i$, this expresses x as a convex combination with every coefficient positive, so x lies in the relative interior of the convex hull. If $0 < |x - a_i| \leq \delta$, then $|x - a_k| \geq |a_i - a_k| - |x - a_i| \geq L_{ik} - \delta > 0$ for $k \neq i$. Splitting off the i th term and using the triangle inequality therefore gives

$$|\nabla V(x)| \geq \frac{q_i}{|x - a_i|^2} - \sum_{k \neq i} \frac{q_k}{|x - a_k|^2} \geq \frac{q_i}{\delta^2} - \sum_{k \neq i} \frac{q_k}{(L_{ik} - \delta)^2} > 0.$$

$\square$

Lemma 6.2 (Global index formula). *Let $q_i \neq 0$, let $Q = \sum_i q_i \neq 0$, and suppose that $\text{Crit}(V)$ is finite. Then*

$$\sum_{x \in \text{Crit}(V)} \text{ind}_x(\nabla V) = \sum_i \text{sgn}(q_i) - \text{sgn}(Q). \quad (16)$$

Proof. Write $F = \nabla V$. Choose pairwise disjoint closed balls about the sites and the critical points, and a large ball containing all of them, so that F is nonzero on every boundary sphere. Give each sphere its standard outward orientation as the boundary of the corresponding ball. On the region obtained from the large ball by deleting the interiors of the small balls, the map $F/|F|$ is defined. The inner spheres have the opposite of their standard orientations as boundary components of this region; consequently, the homological boundary relation gives

$$\deg_{\text{outer}}(F/|F|) = \sum_i \deg_{a_i}(F/|F|) + \sum_{x \in \text{Crit}(V)} \deg_x(F/|F|).$$

Uniformly for $\omega \in S^2$,

$$\nabla V(R\omega) = -\frac{Q}{R^2}\omega + O(R^{-3}),$$

and hence the outer degree is $-\operatorname{sgn}(Q)$. Similarly,

$$\nabla V(a_i + r\omega) = -\frac{q_i}{r^2}\omega + O(1),$$

so the degree on a standard outward-oriented sphere about a_i is $-\operatorname{sgn}(q_i)$. The degree on a sphere about a critical point is its local index. Thus

$$-\operatorname{sgn}(Q) = \sum_{x \in \operatorname{Crit}(V)} \operatorname{ind}_x(\nabla V) - \sum_i \operatorname{sgn}(q_i),$$

which is (16). □

It would still be invalid to infer from an index sum of 3 that there are at least three critical points unless their indices are controlled. In particular, totally degenerate equilibria can have larger index. The next proposition describes the complementary case.

Proposition 6.3 (Index bound away from total degeneracy). *Let x_0 be an isolated critical point of V with $\operatorname{Hess} V(x_0) \neq 0$. Then $|\operatorname{ind}_{x_0}(\nabla V)| \leq 1$.*

Proof. The Hessian is symmetric and traceless. Since it is nonzero, its rank is 2 or 3: a nonzero symmetric matrix of rank one has nonzero trace. In the rank-three case the zero is nondegenerate and has index ± 1 .

In the rank-two case the nonzero eigenvalues are α and $-\alpha$, since the trace vanishes. The splitting lemma [5, Lemma 1] gives a local diffeomorphism $x = \psi(y)$, with $\psi(0) = x_0$, in which

$$(V \circ \psi)(y) = V(x_0) + \frac{\alpha}{2}(y_1^2 - y_2^2) + w(y_3), \quad w(0) = w'(0) = w''(0) = 0.$$

The isolation of x_0 implies that 0 is an isolated zero of w' . The gradient index is unchanged by this coordinate transformation. Indeed, the chain rule gives

$$\nabla(V \circ \psi)(y) = D\psi(y)^T \nabla V(\psi(y)).$$

After shrinking the neighborhood, the straight-line homotopy from $D\psi(y)^T$ to $D\psi(0)^T$ consists of invertible matrices, so it introduces no boundary zero. Precomposition by ψ contributes the factor $\operatorname{sgn} \det D\psi(0)$ to local degree, and postcomposition by $D\psi(0)^T$ contributes the same factor. Their product is 1, so the gradient index is unchanged. The product property of local degree now gives

$$\operatorname{ind}_{x_0}(\nabla V) = \deg_0(\alpha y_1, -\alpha y_2, w'(y_3)) = -\deg_0(w').$$

Choose $\eta > 0$ sufficiently small that w' has no zero in $[-\eta, \eta] \setminus \{0\}$. The one-dimensional degree formula then yields

$$\operatorname{ind}_{x_0}(\nabla V) = -\frac{\operatorname{sgn} w'(\eta) - \operatorname{sgn} w'(-\eta)}{2} \in \{-1, 0, 1\}.$$

□

Proposition 6.3 is not needed for Theorem 1.1, in which every equilibrium is nondegenerate; it is recorded because it shows that only a totally degenerate equilibrium can carry an index of absolute value larger than one. The following example realizes such an equilibrium, and shows why the final perturbation to a Morse potential is necessary.

Example 6.4 (A single equilibrium of index three). Let

$$a_1 = (1, 1, 1), \quad a_2 = (1, -1, -1), \quad a_3 = (-1, 1, -1), \quad a_4 = (-1, -1, 1)$$

carry four unit positive charges. Then the origin is the only critical point:

$$\text{Crit}(V) = \{0\}, \quad \text{Hess } V(0) = 0, \quad \text{ind}_0(\nabla V) = 3.$$

The vertices satisfy

$$\sum_{i=1}^4 a_i = 0, \quad \sum_{i=1}^4 a_i a_i^T = 4I_3, \quad |a_i|^2 = 3.$$

Consequently, direct differentiation gives

$$\nabla V(0) = \sum_{i=1}^4 \frac{a_i}{|a_i|^3} = 0, \quad \text{Hess } V(0) = \sum_{i=1}^4 \left(\frac{3a_i a_i^T}{|a_i|^5} - \frac{I_3}{|a_i|^3} \right) = 0.$$

It remains to prove uniqueness.

Put $s_i = a_i \cdot x$, $\rho = |x|^2$, and $r_i = |x - a_i|$. Since $\sum_i a_i a_i^T = 4I_3$,

$$x = \frac{1}{4} \sum_{i=1}^4 s_i a_i. \tag{17}$$

Because $\sum_i a_i = 0$, the numbers $(1 + s_i)/4$ sum to 1, and by (17) they satisfy $\sum_i \frac{1+s_i}{4} a_i = x$. Hence the barycentric coordinates of x relative to the tetrahedron and the squared distances are

$$\lambda_i = \frac{1 + s_i}{4}, \quad r_i^2 = \rho + 3 - 2s_i.$$

By Lemma 6.1, every equilibrium is in the interior of the tetrahedron, so $\lambda_i > 0$. The vectors $x - a_i$ span $\mathbb{R}^3$ and satisfy the unique relation $\sum_i \lambda_i (x - a_i) = 0$. Consequently

$$\sum_i \frac{x - a_i}{r_i^3} = 0 \iff \lambda_1 r_1^3 = \dots = \lambda_4 r_4^3.$$

Equivalently, since $4\lambda_i r_i^3 = \chi(s_i)$, the four values of

$$\chi(s) = (1 + s)(\rho + 3 - 2s)^{3/2}$$

at $s = s_i$ are equal. Since $\lambda_i > 0$ and $r_i > 0$, all s_i lie in the interval

$$I_\rho = \left(-1, \frac{\rho + 3}{2} \right).$$

Moreover, $\rho/5 \in I_\rho$, and

$$\chi'(s) = (\rho + 3 - 2s)^{1/2}(\rho - 5s).$$

Thus χ is strictly increasing on $(-1, \rho/5)$ and strictly decreasing on $(\rho/5, (\rho + 3)/2)$. A level therefore has at most two preimages, so the s_i take at most two distinct values. If all the s_i are equal, then $\sum_i s_i = 0$ gives $s_i = 0$ and (17) gives $x = 0$. Otherwise the multiplicities are $2 + 2$ or $1 + 3$. In the latter case, after relabeling, $s_2 = s_3 = s_4$, and (17), together with $a_2 + a_3 + a_4 = -a_1$, places x on the vertex axis $\mathbb{R}a_1$. In the former case, after relabeling, $s_1 = s_2$ and $s_3 = s_4$, and the same formula places x on the axis through the midpoints of the opposite edges $a_1 a_2$ and $a_3 a_4$.

The tetrahedral symmetry is transitive on the four vertex axes and on the three opposite-edge axes, so it suffices to treat one representative of each type. The edge-midpoint axis through the midpoints of $a_1 a_2$ and $a_3 a_4$ is the x -axis, and its intersection with the open tetrahedron is $\{(w, 0, 0) : -1 < w < 1\}$; the tetrahedral symmetry $(x, y, z) \mapsto (-x, -y, z)$ reverses it, so a noncentral point on it may be written $x = (w, 0, 0)$ with $0 < w < 1$. There $r_1 = r_2 =$

$(w^2 - 2w + 3)^{1/2}$ and $r_3 = r_4 = (w^2 + 2w + 3)^{1/2}$, and the second and third components of ∇V vanish, so the equilibrium equation reduces to

$$\frac{w - 1}{(w^2 - 2w + 3)^{3/2}} + \frac{w + 1}{(w^2 + 2w + 3)^{3/2}} = 0.$$

The first summand is negative and the second is positive. After moving the first summand to the other side, both sides are positive, so squaring is reversible. Clearing denominators then gives the equivalent condition

$$8w(7 - 2w^2 - w^4) = 0,$$

which is impossible for $0 < w < 1$ because $7 - 2w^2 - w^4 > 4$.

On a vertex axis write $x = ca_1$, where $-1/3 < c < 1$. There $r_1 = \sqrt{3}(1 - c)$ and $r_2 = r_3 = r_4 = (3c^2 + 2c + 3)^{1/2}$; by the threefold symmetry about this axis, ∇V is parallel to a_1 , so taking the inner product with a_1 shows that the equilibrium equation is equivalent to

$$\frac{3c + 1}{(3c^2 + 2c + 3)^{3/2}} = \frac{1}{3\sqrt{3}(1 - c)^2}.$$

Both sides are positive on this interval, so squaring is reversible. Clearing denominators and putting $u = 1 - c \in (0, 4/3)$ gives the equivalent condition

$$8c^2 P(u) = 0, \quad P(u) = 27u^4 - 72u^2 + 64u - 64.$$

But

$$P(u) \leq -8(3u^2 - 8u + 8) < 0 \quad (0 < u < 4/3),$$

where the first inequality is equivalent to $3u^2(9u^2 - 16) \leq 0$ and the quadratic in parentheses has negative discriminant. Hence $c = 0$, so there is no noncentral equilibrium on a vertex axis. Finally, Lemma 6.2 gives $\text{ind}_0(\nabla V) = 4 - 1 = 3$.

Lemma 6.5 (Generic nondegeneracy). *Let $a_1, \dots, a_4$ be affinely independent. Then, for almost every $q \in \mathbb{R}^4$, every critical point of V_q is nondegenerate.*

Proof. Set

$$\Phi : (\mathbb{R}^3 \setminus \{a_i\}) \times \mathbb{R}^4 \longrightarrow \mathbb{R}^3, \quad \Phi(x, q) = \nabla V_q(x).$$

Its partial derivative in the charge variables is

$$c \longmapsto - \sum_i c_i \frac{x - a_i}{|x - a_i|^3}.$$

The vectors $x - a_i$ span $\mathbb{R}^3$ for every x , because the differences $(x - a_i) - (x - a_4) = a_4 - a_i$ span $\mathbb{R}^3$. The columns of $D_q \Phi$ are nonzero scalar multiples of these vectors. Thus $D_q \Phi$ is surjective, 0 is a regular value of Φ , and $W = \Phi^{-1}(0)$ is a smooth four-manifold.

Let $\pi : W \rightarrow \mathbb{R}^4$ be projection to the charge vector. At $(x, q) \in W$,

$$T_{(x,q)} W = \{(\xi, \eta) : \text{Hess } V_q(x)\xi + D_q \Phi(x, q)\eta = 0\}.$$

Since both W and the target of π have dimension four, $d\pi$ is surjective exactly when its kernel is zero. That kernel consists of the $(\xi, 0)$ with $\text{Hess } V_q(x)\xi = 0$. Hence q is a regular value of π exactly when every critical point of V_q is nondegenerate. Sard's theorem shows that the regular charge vectors have full measure in $\mathbb{R}^4$; in particular, they are dense. $\square$

Proof of Theorem 1.1. Take $t = 1/2$. Then

$$H = 3, \quad p_0 = 3\sqrt{5}, \quad x_0 = (0, 0, 1/2), \quad A = \frac{144\sqrt{5}}{625}, \quad \kappa = \frac{7}{25}.$$

Throughout, charge vectors are indexed by $0, 1, 2, 3$ so as to match the sites $b_0, \dots, b_3$ of (1), and $V_{q''}$ denotes the potential of the charges q'' at these sites.

Apply Lemma 5.2 with $t = 1/2$, and let r_0 and ε_0 be the constants provided there. Choose $0 < \varepsilon < \varepsilon_0$, and let $q = (1, 1, 1, 3\sqrt{5} - \varepsilon)$. All four charges are positive. The lemma supplies eight nondegenerate critical points in $B(x_0, r_0)$, and the degree of the gradient on that ball is zero.

We next obtain a compact confinement set that is uniform for nearby charge vectors. Choose a bounded convex open neighborhood U of q whose closure is contained in the positive orthant. There are constants $0 < m_i \leq q''_i \leq M_i$ for $q'' \in U$ and $0 \leq i \leq 3$. Because the sites are fixed and distinct, one can choose $0 < \delta < \frac{1}{2} \min_{i \neq k} |b_i - b_k|$ so small that, for $i = 0, 1, 2, 3$,

$$\frac{m_i}{\delta^2} > \sum_{k \neq i} \frac{M_k}{(|b_i - b_k| - \delta)^2}.$$

Applying Lemma 6.1 with $L_{ik} = |b_i - b_k|$ confines every critical point of every $V_{q''}$, $q'' \in U$, to the compact set

$$K = \text{conv}\{b_0, b_1, b_2, b_3\} \cap \bigcap_{i=0}^3 \{x : |x - b_i| \geq \delta\},$$

which is disjoint from the sites.

Let $z_1, \dots, z_8$ be the eight nondegenerate critical points supplied by Lemma 5.2. Choose pairwise disjoint neighborhoods of these points whose closures lie in $B(x_0, r_0)$. The parameter-dependent implicit function theorem gives an open neighborhood $U_1 \subset U$ of q such that each $V_{q''}$, $q'' \in U_1$, has a distinct critical point in each of these eight neighborhoods.

Set $\Sigma = \partial B(x_0, r_0)$ and

$$\mu = \min_{x \in \Sigma} |\nabla V_q(x)| > 0, \quad C_\Sigma = \max_{x \in \Sigma} \sum_{i=0}^3 \frac{1}{|x - b_i|^2} < \infty.$$

After shrinking U_1 , assume that $\|q'' - q\|_\infty C_\Sigma < \mu$ for every $q'' \in U_1$. Since $b_0, \dots, b_3$ are affinely independent, the density conclusion in Lemma 6.5, applied to these four sites, allows us to choose $q' \in U_1$ such that every critical point of $V_{q'}$ is nondegenerate. All these critical points lie in K . Their set is closed in K , hence compact, and every point of it is isolated; it is therefore finite.

For $0 \leq s \leq 1$, put $q_s = (1 - s)q + sq'$. Linearity of the field in the charge vector gives, for $x \in \Sigma$,

$$|\nabla V_{q_s}(x)| \geq |\nabla V_q(x)| - s\|q' - q\| * \infty C * \Sigma > 0.$$

Thus the straight charge-vector homotopy is boundary-zero-free, and homotopy invariance preserves the local degree:

$$\deg(\nabla V_{q'}, B(x_0, r_0), 0) = 0.$$

The eight continued critical points are distinct and lie inside this ball. Since all critical points of $V_{q'}$ are nondegenerate, additivity of degree shows that the critical points in the ball have total index zero and number at least eight.

Finally, all q'_i and their sum are positive, so Lemma 6.2 gives

$$\sum_{x \in \text{Crit}(V_{q'})} \text{ind}_x(\nabla V_{q'}) = 4 - 1 = 3.$$

Taking $s = 1$ in the previous display shows that $\nabla V_{q'}$ does not vanish on Σ , so every critical point of $V_{q'}$ lies either inside or outside $B(x_0, r_0)$. The critical points outside $B(x_0, r_0)$ therefore have indices summing to 3. Each is nondegenerate and has index ± 1 , so there are at least three of them. Together with the eight inside the ball, this gives

$$\# \text{Crit}(V_{q'}) \geq 8 + 3 = 11.$$

The sites are affinely independent, the charges are positive, and the critical set is finite, which proves the theorem. $\square$

Remark 6.6. No numerical certification is used. The seed and local-model calculations are exact, while the existence of sufficiently small perturbations follows from analytic estimates, the implicit function theorem, and compactness. The final Morse charge vector q' is selected by Sard's theorem rather than specified numerically.

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